Module 5 Inference in FOL – Forward and Backward Chaining

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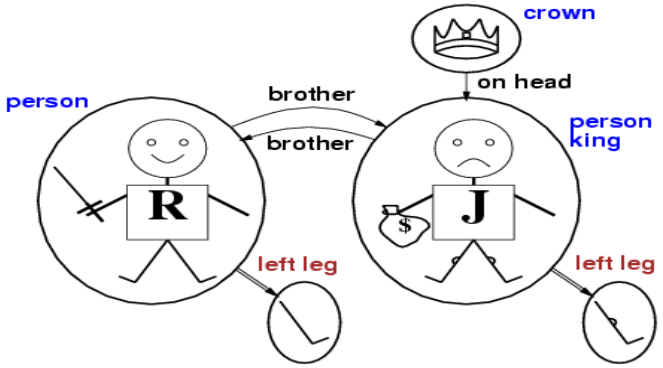
Outline

- Forward Chaining
- Backward Chaining

Inference Rules



- And-Elimination (AE)

$$Crown(C_1) \wedge OnHead(C_1,John)$$

Can be written as two clauses

- $Crown(C_1)$
- $OnHead(C_1,John)$

- And-Introduction (AI)
  - $Crown(C_1)$
  - $OnHead(C_1,John)$

Can be written as

$$Crown(C_1) \wedge OnHead(C_1,John)$$

Inference Rules

- Modus Ponens (MP)/Implication-Elimination

A => B, A

B

Example

Fish(George)->Swims(George)

Fish(George)

Swims(George)

Inference Rules

- Modus Ponens (MP)/Implication-Elimination

A => B, A

B

Example

Fish(George)->Swims(George)

Fish(George)

Swims(George)

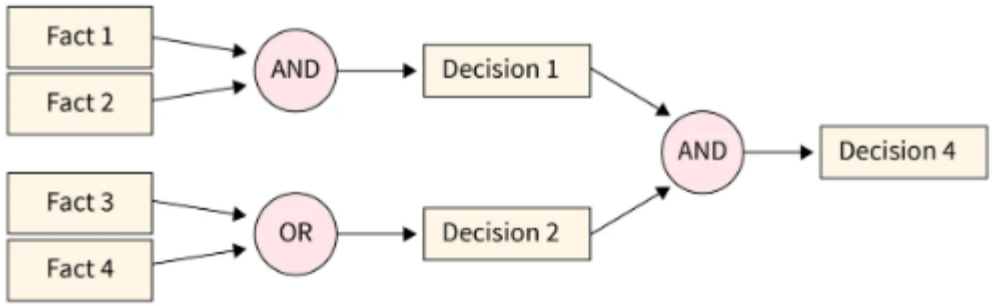
Forward chaining algorithm

```
function FOL-FC-ASK(KB, α) returns a substitution or false
   repeat until new is empty
      new ← { }
      for each sentence r in KB do
         ( p1 ∧ ... ∧ pn ⇒ q) ← STANDARDIZE-APART(r)
         for each θ such that (p1 ∧ ... ∧ pn)θ = (p'1 ∧ ... ∧ p'n)θ
                        for some p'1, ..., p'n in KB
            q' ← SUBST(θ, q)
            if q' is not a renaming of a sentence already in KB or new then do
               add q' to new
               φ ← UNIFY(q', α)
               if φ is not fail then return φ
      add new to KB
   return false
```

Forward Chaining

- Idea: Starting from the known facts, it triggers all the rules whose premises are satisfied, adding their conclusions to the known facts.
- The process repeats until the query is answered or no new facts are added.
- i.e.Fire any rule whose premises are satisfied in the KB, add its conclusion to the KB, until query is found.

Forward Chaining



- **Data-Driven**
- **Bottom-Up Approach**
- **Breadth-First Search Strategy**
- **Possibility of Irrelevant Rules**

Example 1

- Rimi is hungry
- if Rimi is hungry, she barks.
- if Rimi is barking then Raja is angry.

Convert the above sentence in FOL. Apply the forward chaining algorithm to prove Raja is Angry.

Example 1 (contd…)

- Rimi is hungry — Hungry(Rimi) …. 1
- if Rimi is hungry, she barks. — Hungry(Rimi)→Barks(Rimi)….2
- if Rimi is barking then Raja is angry. — Barks(Rimi)→Angry(Raja))…3
- Prove Raja is Angry

  Angry(Raja)
- Convert in FOL

Angry(Raja) From 3

Barks(Rimi) From 2

Hungry(Rimi) KB Predicate

Example 2

- The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.

- Prove that Col. West is a criminal using Forward Chaining Algorithm

Example 2 (contd…)

... it is a crime for an American to sell weapons to hostile nations:

$American(x) \wedge Weapon(y) \wedge Sells(x,y,z) \wedge Hostile(z) \Rightarrow Criminal(x)$ ...1

Nono … has some missiles, i.e.,

$\exists x\ Owns(Nono,x) \wedge Missile(x)$:

$Owns(Nono,M_1)$ and $Missile(M_1)$ ...2

… all of its missiles were sold to it by Colonel West

$Missile(x) \wedge Owns(Nono,x) \Rightarrow Sells(West,x,Nono)$ ...3

Example 2 (contd...)

... Missiles are weapons:

$Missile(x) \Rightarrow Weapon(x) \quad ...4$

An enemy of America counts as "hostile“:

$Enemy(x,America) \Rightarrow Hostile(x) \quad ...5$

West, who is American …

$American(West) \quad ...6$

The country Nono, an enemy of America …

$Enemy(Nono,America) \quad ...7$

Example 2 (contd…)

American(West)

Missile(M1)

Owns(Nono,M1)

Enemy(Nono,America)

American(x) $\wedge$ Weapon(y) $\wedge$ Sells(x,y,z) $\wedge$ Hostile(z) $\Rightarrow$ Criminal(x)

Owns(Nono,M_1) $\wedge$ Missile(M_1)

Missile(x) $\wedge$ Owns(Nono,x) $\Rightarrow$ Sells(West,x,Nono)

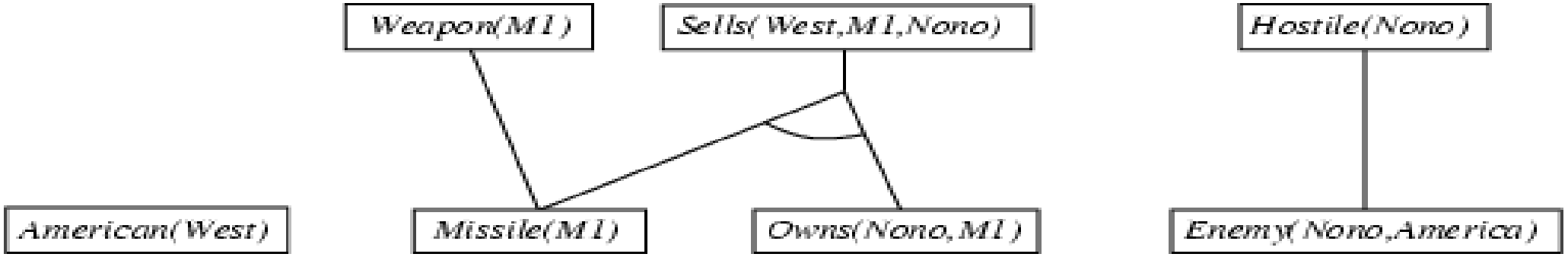
Missile(x) $\Rightarrow$ Weapon(x)

American(West)

Enemy(x,America) $\Rightarrow$ Hostile(x)

Enemy(Nono,America)

Example 2 (contd…)



American(x) $\wedge$ Weapon(y) $\wedge$ Sells(x,y,z) $\wedge$ Hostile(z) $\Rightarrow$ Criminal(x)

Owns(Nono,M_1) $\wedge$ Missile(M_1)

Missile(x) $\wedge$ Owns(Nono,x) $\Rightarrow$ Sells(West,x,Nono)

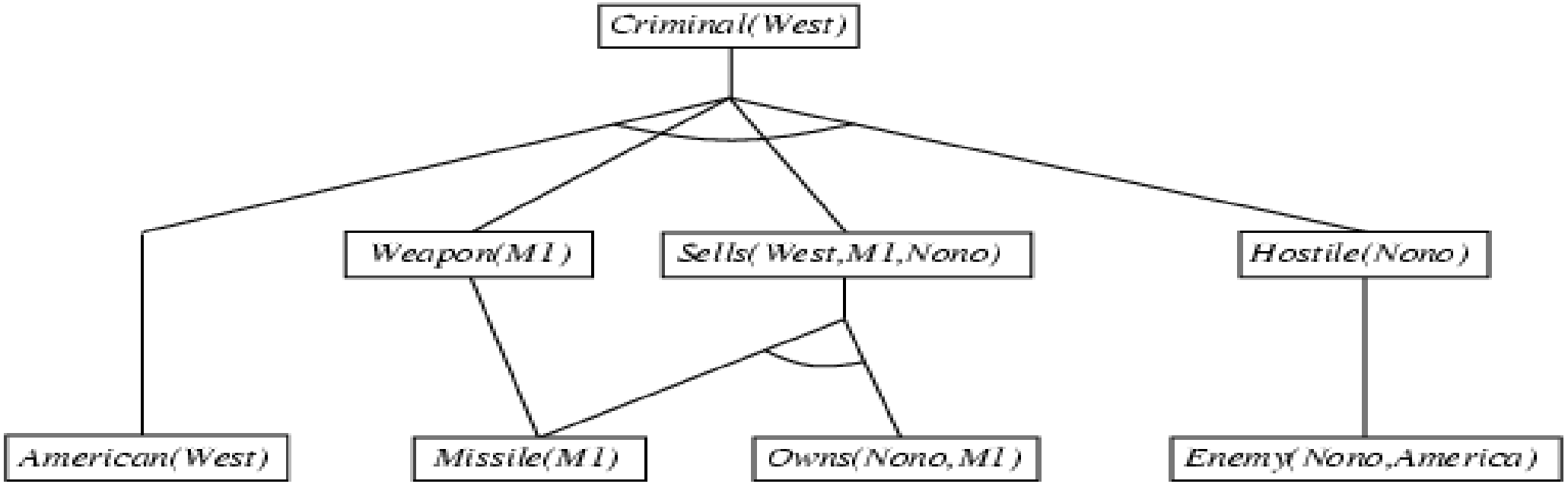
Missile(x) $\Rightarrow$ Weapon(x)

American(West)

Enemy(x,America) $\Rightarrow$ Hostile(x)

Enemy(Nono,America)

Example 2 (contd…)



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Missile(x) $\Rightarrow$ Weapon(x)

American(West)

Enemy(x,America) $\Rightarrow$ Hostile(x)

Enemy(Nono,America)

Example 3

If the maid stole the jewellery, then the butler wasn't guilty. Either the maid stole the jewellery or she milked the cows. If the maid milked the cows, then the butler got the cream.

Goal: Therefore, if the butler was guilty, then he got the cream.

Example 1

If the maid stole the jewellery, then the butler wasn't guilty. Either the maid stole the jewellery or she milked the cows. If the maid milked the cows, then the butler got the cream.

Goal: Therefore, if the butler was guilty, then he got the cream.

Define the predicates

- Stole(m) → "The maid stole the jewellery"
- Guilty(b) → "The butler is guilty"
- Milk(m) → "The maid milked the cows"
- Cream(b) → "The butler got the cream"

The domain consists of **individuals**:

- m represents the **maid**.
- b represents the **butler**.

G(M) : The maid stole jewelry
G(B) : Butler was guilty
Cows(M) : Maid milked cow
Cream(B) : Butler got cream.

If the maid stole the jewelry, then the butler wasn't guilty.

G(M) -> ~G(B) …1

Either the maid stole the jewelry or she milked the cows.

G(M) v Cows(M) …2

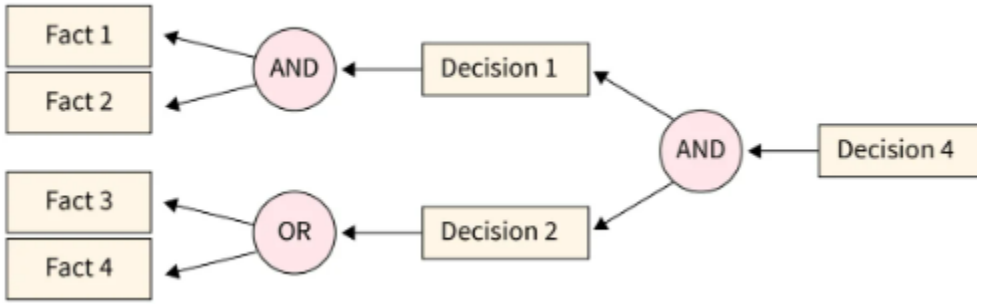
If the maid milked the cows, then the butler got the cream

Cows(M) -> Cream(B) …3

Goal: Therefore, if the butler was guilty, then he got the cream.

G(B) -> Cream(B) …4

Backward Chaining



- **Goal-Driven**
- **Top-Down Approach**
- **Depth-First Search Strategy**
- **Possibility of Infinite Loops**

Example 1

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- Prove Raja is Angry

  Angry(Raja)
- Convert in FOL

Angry(Raja)

From 3

Barks(Rimi)

From 2

Hungry(Rimi)

Example 2

- The law says that it is a crime for an American to sell weapons to hostile nations. The country Nono, an enemy of America, has some missiles, and all of its missiles were sold to it by Colonel West, who is American.

- Prove that Col. West is a criminal. Apply the Backward Chaining Algorithm.

Example 2 (contd…)

... it is a crime for an American to sell weapons to hostile nations:

$American(x) \wedge Weapon(y) \wedge Sells(x,y,z) \wedge Hostile(z) \Rightarrow Criminal(x)$ ...1

Nono … has some missiles, i.e.,

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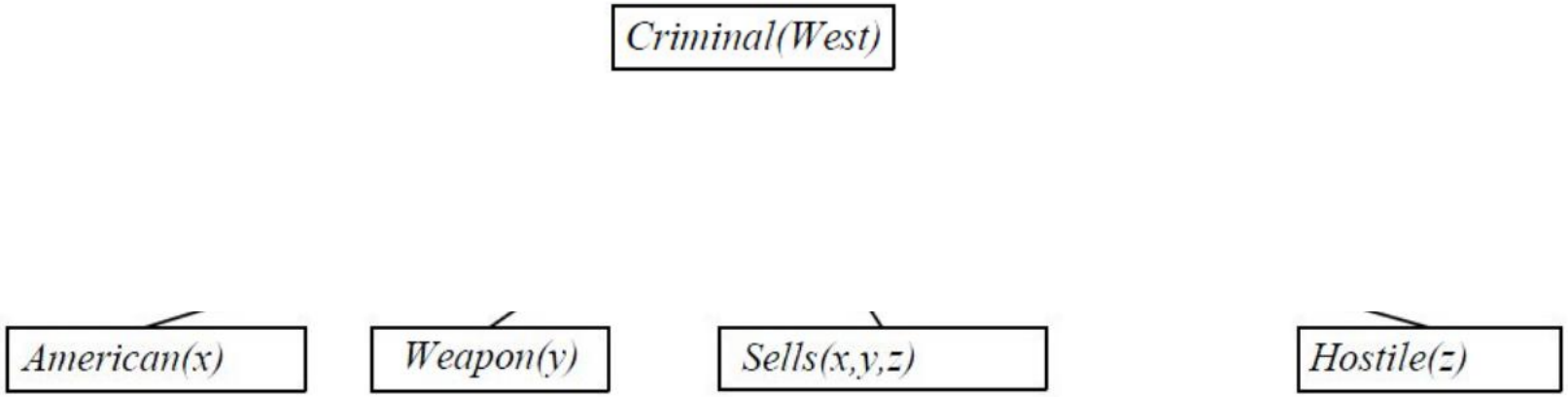


American(x) $\wedge$ Weapon(y) $\wedge$ Sells(x,y,z) $\wedge$ Hostile(z) $\Rightarrow$ Criminal(x)

Owns(Nono,M_1) $\wedge$ Missile(M_1)

Missile(x) $\wedge$ Owns(Nono,x) $\Rightarrow$ Sells(West,x,Nono)

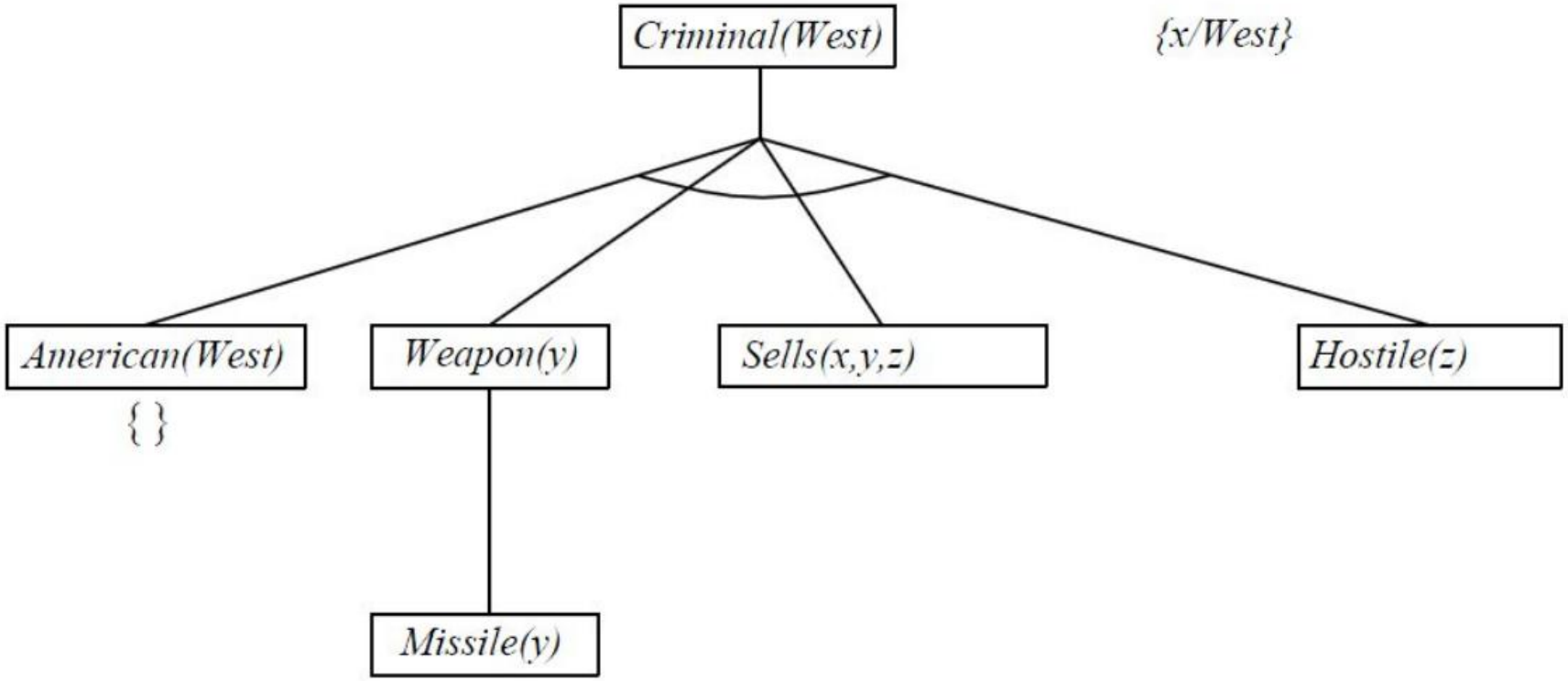
Missile(x) $\Rightarrow$ Weapon(x)

American(West)

Enemy(x,America) $\Rightarrow$ Hostile(x)

Enemy(Nono,America)

Example 2 (contd…)



American(x) $\wedge$ Weapon(y) $\wedge$ Sells(x,y,z) $\wedge$ Hostile(z) $\Rightarrow$ Criminal(x)

Owns(Nono,M_1) $\wedge$ Missile(M_1)

Missile(x) $\wedge$ Owns(Nono,x) $\Rightarrow$ Sells(West,x,Nono)

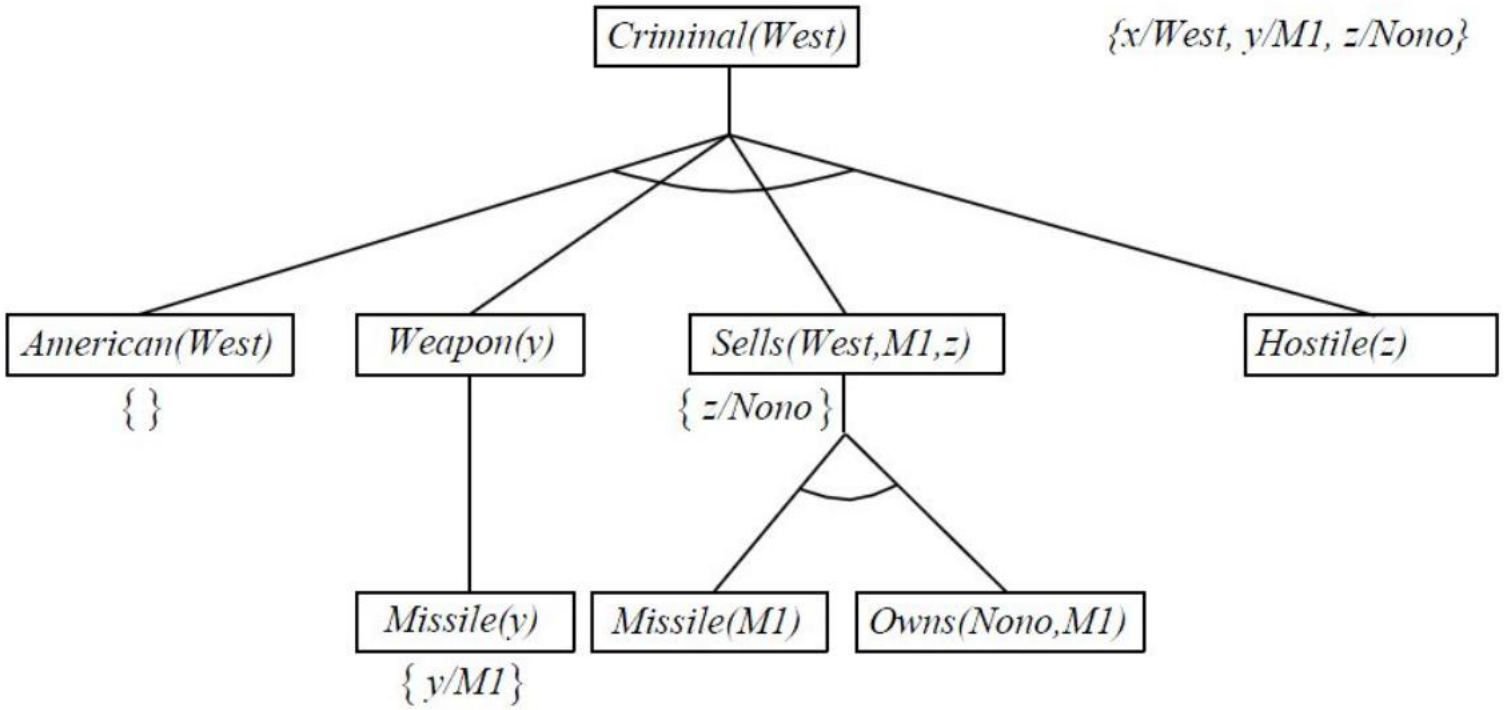
Missile(x) $\Rightarrow$ Weapon(x)

American(West)

Enemy(x,America) $\Rightarrow$ Hostile(x)

Enemy(Nono,America)

Example 2 (contd…)



American(x) ∧ Weapon(y) ∧ Sells(x,y,z) ∧ Hostile(z) ⇒ Criminal(x)

Enemy(Nono,America)

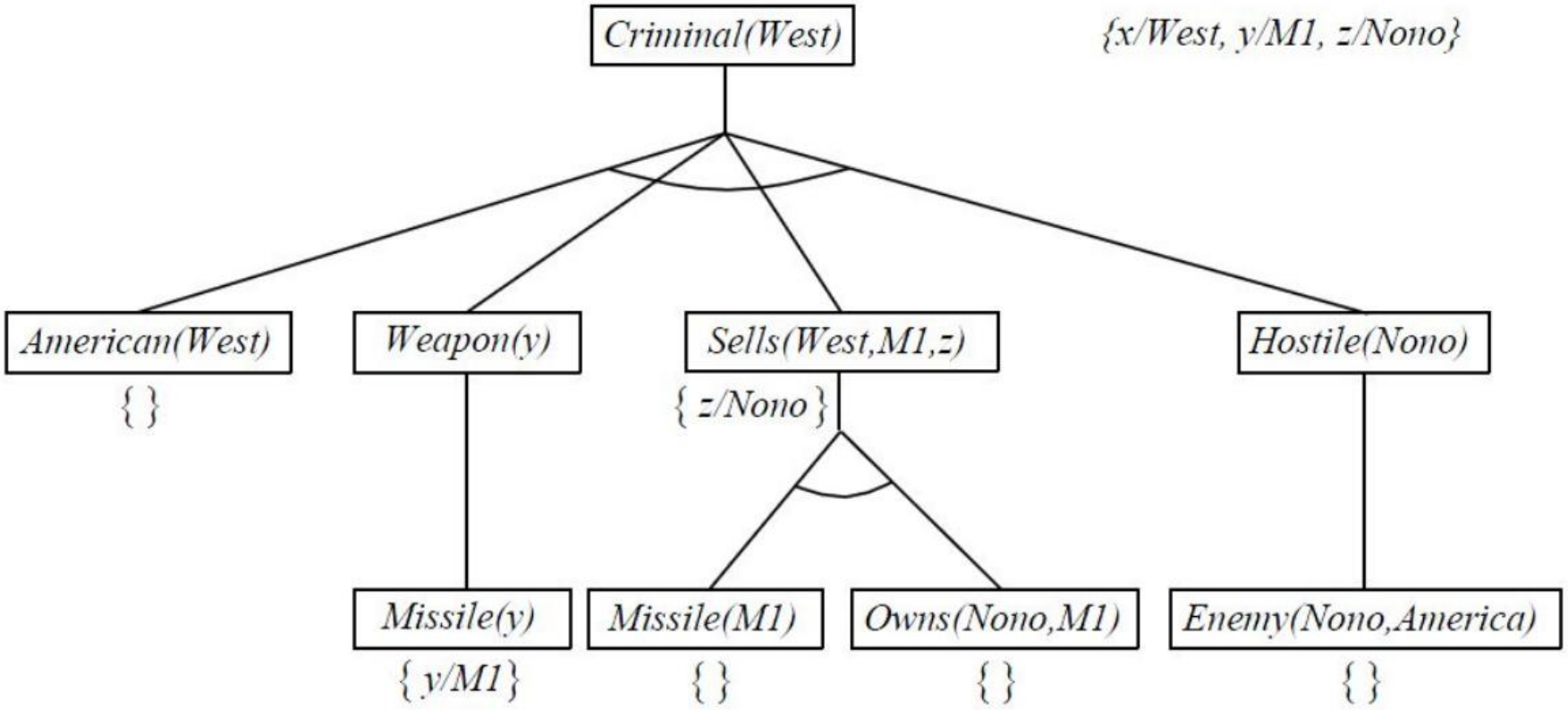
Owns(Nono,M_1) ∧ Missile(M_1)

Missile(x) ∧ Owns(Nono,x) ⇒ Sells(West,x,Nono)

Missile(x) ⇒ Weapon(x)

Enemy(x,America) ⇒ Hostile(x)

Example 2 (contd…)



Practice

- Translate the following set of sentences into FOL. Use Forward and Backward chaining to prove if Marcus was loyal to Caesar.
  - Marcus was a man.
  - Marcus was a Pompeian.
  - All Pompeian were Romans.
  - Caesar was a ruler.
  - All Romans were either loyal to Caesar or hated him.
  - Everyone is loyal to someone.
  - People only try to assassinate rulers they aren't loyal to.
  - Marcus tried to assassinate Caesar.

Forward Vs Backward Chaining

	Forward Chaining	Backward Chaining
Approach	Data-Driven: Starts from facts and applies rules to reach conclusions.	Goal-Driven: Starts with a goal and works backward to verify if facts support it.
Search Strategy	Breadth-First Search: Explores multiple rules at the same level.	Depth-First Search: Focuses deeply on one path before trying others.
Direction	Bottom-Up: Moves from facts to conclusions.	Top-Down: Begins with the goal and works towards the facts.
Efficiency	May explore irrelevant rules, potentially reducing efficiency.	Risk of getting stuck in infinite loops if not managed properly.

Forward Vs Backward Chaining

	Forward Chaining	Backward Chaining
Memory Usage	Can require more memory as it processes multiple rules at once.	More memory-efficient as it focuses on specific goals or sub-goals.
Complexity	Easier to implement for systems with many rules and data.	Can be more complex due to recursive searches for supporting facts.
Performance	Performs better when all relevant data is known upfront.	Works well when only specific information or goals are of interest.
Examples of Use Cases	Troubleshooting, diagnostics, and prediction systems.	Query systems, expert systems, and decision-making models.

Question ?